# CE205 Databases and Information Retrieval

# Indexing in Information Retrieval System

Class 7

## Objective

The purpose of this exercise is to construct data structures for information retrieval system, processing of queries, and creation of a bag-of-words model.

## Problem 1

**Construct term-document incidence matrix for the following collection of documents**

**Doc 1** new home sales top forecasts

**Doc 2** home sales rise in july

**Doc 3** increase in home sales in july

**Doc 4** july new home sales rise

| | D1 | D2 | D3 | D4 |
|---|---|---|---|---|
| forecast | | | | |
| home | | | | |
| in | | | | |
| Increase | | | | |
| july | | | | |
| new | | | | |
| rise | | | | |
| sales | | | | |
| top | | | | |

## Problem 2:

**Consider the given term-document incidence matrix.**

| DOC / Term | $D_1$ | $D_2$ | $D_3$ | $D_4$ | $D_5$ |
|---|---|---|---|---|---|
| a | 1 | 1 | 1 | 0 | 1 |
| b | 0 | 1 | 0 | 1 | 1 |
| c | 0 | 0 | 1 | 0 | 1 |

**What document(s) will be retrieved by the following queries?**

Q1 = a AND (b AND (NOT c))

Q 2= a AND (b OR (NOT c))

**Solution for Q1**

**Solution for Q2**

## Problem 3

**Construct inverted index for the following collection of documents**

**Doc 1** new home sales top forecasts

**Doc 2** home sales rise in july

**Doc 3** increase in home sales in july

**Doc 4** july new home sales rise

*Inverted index should contain terms, document frequencies and postings lists.* *(see slide 27)*

**SOLUTION: (complete the table)**

| Terms | Document Frequency | Posting list |
|---|---|---|
| forecast | | |
| home | | |
| in | | |
| Increase | | |
| july | | |
| new | | |
| rise | | |
| sales | | |
| top | | |

## Problem 4

**Construct positional inverted index for the following collection of documents**

**Doc 1** new home sales top forecasts

**Doc 2** home sales rise in july

**Doc 3** increase in home sales in july

**Doc 4** july new home sales rise

*Use collection frequency, and also mention term frequency for each document. (see slide 34 and 35)*

**SOLUTION: (complete the table)**

| | Collection Frequency | Positional Inverted Index |
|---|---|---|
| forecast | | |
| home | | |
| in | | |
| Increase | | |
| july | | |
| new | | |
| rise | | |
| sales | | |
| top | | |

## Problem 5

### Consider the following positional postings lists:

| | |
|---|---|
| learning, 5 | $\langle d_2, 1 : [52]; d_4, 3 : [35, 105, 148]; d_5, 1 : [78] \rangle$ |
| and, 8 | $\langle d_1, 1 : [63]; d_2, 3 : [68, 124, 143]; d_3, 1 : [39]; d_4, 2 : [44, 114]; d_5, 1 : [47] \rangle$ |
| teaching, 1 | $\langle d_4, 1 : [48] \rangle$ |

**What will be the result of the query learning and teaching?**

**SOLUTION:**

## Problem 6

**Construct a bag-of-words model for the following collection.**

**Doc 1** **it was the best of times**

**Doc 2** **it was the worst of times**

**Doc 3** **it was the age of wisdom not foolishness**

**Doc 4** **it was the age of foolishness not wisdom**

**Solution guideline:**

1. **First, determine the unique terms in the collection.**
2. **Then, determine document vectors**

**Bag-of-words Model:**

| | **Terms** | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | |
| **Doc1** | | | | | | | | | | | |
| **Doc2** | | | | | | | | | | | |
| **Doc3** | | | | | | | | | | | |
| **Doc4** | | | | | | | | | | | |